1. [10 points] **Information Filter:** In this problem you will derive an equivalent form of the Kalman filter called the information filter, which is advantageous for some problems. Consider the quantities $\Omega_{t|t} := \Sigma_{t|t}^{-1}$, $\Omega_{t|t-1} := \Sigma_{t|t-1}^{-1}$, called the information matrix, and $\xi_{t|t} := \Sigma_{t|t}^{-1}\mu_{t|t}$, $\xi_{t|t-1} := \Sigma_{t|t-1}^{-1}\mu_{t|t-1}$ called the information state.

   (a) Derive recursive expressions for $\Omega_{t|t}$ and $\Omega_{t|t-1}$ from the Kalman filter equations. Your expressions should each require, at most, a single matrix inversion (aside from R_t^{-1} and Q_t^{-1}, which can be precomputed). The following matrix inversion identity will be useful: $(A+UCV)^{-1} = A^{-1} - A^{-1}U(VA^{-1}U + C^{-1})^{-1}VA^{-1}$ for matrices A, U, C, and V.

   (b) Similarly, derive recursive expressions for $\xi_{t|t}$ and $\xi_{t|t-1}$ from the Kalman filter equations. Again, your expressions should require no more than a single matrix inverse except for R_t^{-1} and Q_t^{-1}. The following matrix identity will be useful: $(A+UCV)^{-1}UC = A^{-1}U(C^{-1}+VA^{-1}U)^{-1}$.

   (c) Matrix inversion dominates the computational complexity of the Kalman filter equations (with larger matrices taking longer to invert). With this in mind, when might one prefer the standard form or the information form of the Kalman filter?

a) $(A+UCV)^{-1} = A^{-1} - A^{-1}U(VA^{-1}U + C^{-1})^{-1}VA^{-1}$

$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$ $\rightarrow$ $A=U$ $A^T=V$ $Q=A$

$\Omega_{t|t-1} = (A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1})^{-1} = Q_{t-1}^{-1} - Q_{t-1}^{-1}A_{t-1}(A_{t-1}^T Q_{t-1}^{-1} A_{t-1} + \Sigma_{t-1|t-1}^{-1})^{-1}A_{t-1}^T Q_{t-1}^{-1}$ ($\Omega_{t-1|t-1}$)

$\Omega_{t|t-1} = Q_{t-1}^{-1} - Q_{t-1}^{-1}A_{t-1}(A_{t-1}^T Q_{t-1}^{-1} A_{t-1} + \Omega_{t-1|t-1})^{-1}A_{t-1}^T Q_{t-1}^{-1}$

$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1}C^T(C\Sigma_{t|t-1}C^T + R)^{-1}C\Sigma_{t|t-1}$

$\Sigma_{t|t-1} = A^{-1}$ $\rightarrow$ $\Sigma_{t|t-1}^{-1} = A$ $C^T = U$ $C = V$ $R = C^{-1}$ $\rightarrow$ $R^{-1} = C$

$\Sigma_{t|t} = (\Sigma_{t|t-1}^{-1} + C^T R^{-1} C)^{-1}$ $\therefore$ $\Sigma_{t|t}^{-1} = \Sigma_{t|t-1}^{-1} + C^T R^{-1} C$

$\therefore$ $\Omega_{t|t} = \Omega_{t|t-1} + C_t^T R_t^{-1} C_t$

b) $(A+UCV)^{-1}UC = A^{-1}U(C^{-1}+VA^{-1}U)^{-1}$ $Q=A$ $\Sigma=C$ $A=U$ $A^T=V$

$(A+UCV)^{-1}U = A^{-1}U(C^{-1}+VA^{-1}U)^{-1}C^{-1}$

$\xi_{t|t-1} = \Sigma_{t|t-1}^{-1}\mu_{t|t-1} = \Sigma_{t|t-1}^{-1}(A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1})$ $\leftarrow \mu_{t-1|t-1} = \Sigma_{t-1|t-1}\xi_{t-1|t-1}$

$= \Sigma_{t|t-1}^{-1}A_{t-1}\mu_{t-1|t-1} + \Sigma_{t|t-1}^{-1}B_{t-1}u_{t-1}$

$= (A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1})^{-1}A_{t-1}\mu_{t-1|t-1} + \Omega_{t|t-1}B_{t-1}u_{t-1}$

$= Q_{t-1}^{-1}A_{t-1}(\Sigma_{t-1|t-1}^{-1} + A_{t-1}Q_{t-1}^{-1}A_{t-1})^{-1}\Sigma_{t-1|t-1}^{-1}\mu_{t-1|t-1} + \Omega_{t|t-1}B_t u_t$

$\mu_{t|t-1} = Q_{t-1}^{-1}A_{t-1}(\Omega_{t-1|t-1} + A_{t-1}Q_{t-1}^{-1}A_{t-1})^{-1}\xi_{t-1|t-1} + \Omega_{t|t-1}B_t u_t$

$\mu_{t|t} = \mu_{t|t-1} + \Sigma_{t|t-1}C^T(C\Sigma_{t|t-1}C^T + R)^{-1}(y_t - C\mu_{t|t-1})$

$\Sigma_{t|t-1} = A^{-1}$ $\rightarrow$ $\Sigma_{t|t-1}^{-1} = A$ $C^T = U$ $C = V$ $R = C^{-1}$ $\rightarrow$ $R^{-1} = C$

$$\mu_{t|t} = \mu_{t|t-1} + \underbrace{\left(\Sigma_{t|t-1}^{-1} + C^T R^{-1} C\right)^{-1}}_{\Sigma_{t|t}} C^T R^{-1}\left(y - C\mu_{t|t-1}\right)$$

$$\Sigma_{t|t}^{-1}\mu_{t|t} = \Sigma_{t|t}^{-1}\mu_{t|t-1} + C^T R^{-1} y - C^T R^{-1} C \mu_{t|t-1}$$

$$= \left(\Sigma_{t|t}^{-1} - \underbrace{C^T R^{-1} C}_{\Sigma_{t|t-1}^{-1}}\right)\mu_{t|t-1} + C^T R^{-1} y = \Sigma_{t|t-1}\mu_{t|t-1} + C^T R^{-1} y$$

$$\boxed{\xi_{t|t} = \xi_{t|t-1} + C_t^T R_t^{-1} y_t}$$

c) In this filter, to recover the state, $\Sigma_{t|t}$ must be inverted. This is a $n \times n$ matrix. In the original kalman filter, the $m \times m$ quantity of $\left(C\Sigma_{t|t-1}C^T + R\right)^{-1}$ is inverted. Therefore, if $m > n$, it would be more costly to run the original kalman filter, and the information filter would be computationally more efficient

2. [10 points] **Linearizing Dynamics:** Consider the rotational motion of a satellite, which can be modeled by Euler's equations of rigid body motion,

$$J_x\dot{\omega}_x = (J_y - J_z)\omega_y\omega_z + \tau_x$$
$$J_y\dot{\omega}_y = (J_z - J_x)\omega_z\omega_x + \tau_y$$
$$J_z\dot{\omega}_z = (J_x - J_y)\omega_x\omega_y + \tau_z,$$

where $[\omega_x \quad \omega_y \quad \omega_z]^T$ is the angular velocity vector, $J = \text{diag}([J_x \quad J_y \quad J_z])$ is the inertia tensor in principal axes, and $\tau = [\tau_x \quad \tau_y \quad \tau_z]^T$ is the input torque from thrusters.

(a) Use a first order discrete time approximation to find a difference equation model for the satellite. Use the time step $\delta t = 0.001$s.

(b) Find the Jacobian of the resulting nonlinear discrete-time dynamics.

(c) Suppose the satellite gets angular velocity measurements from on-board gyroscopes with the measurement equation

$$y_t = \text{sat}(\omega_t) + v_t,$$

where $\text{sat}(\cdot)$ is the element-wise saturation function defined as follows. Let $\omega_t = [\omega_t^1 \quad \omega_t^2 \quad \omega_t^3]^T$. Then $\text{sat}(\cdot)$ returns a vector with ith element ω_t^i if $|\omega_t^i| \leq c$ for some threshold value c, and $\text{sign}(\omega_t^i)c$ if $|\omega_t^i| > c$ is above the threshold (where $\text{sign}(\cdot)$ returns 1 or -1 depending on the sign of its argument). Find the Jacobian of the measurement function.

a)

$$\dot{J\omega} = \begin{pmatrix} (J_y - J_z)\omega_y\omega_z \\ (J_z - J_x)\omega_z\omega_x \\ (J_x - J_y)\omega_x\omega_y \end{pmatrix} + \tau$$

$$\omega_{t+1} = \omega_t + \delta t\,\dot{\omega}$$

$$\omega_{t+1} = \omega_t + \delta t\, J^{-1}\begin{pmatrix} (J_y - J_z)\omega_y\omega_z \\ (J_z - J_x)\omega_z\omega_x \\ (J_x - J_y)\omega_x\omega_y \end{pmatrix} + \delta t J^{-1}\tau$$

$$\omega_{t+1} = \omega_t + \delta t \begin{bmatrix} \frac{1}{J_x}(J_y - J_z)\,\omega_y\omega_z \\ \frac{1}{J_y}(J_z - J_x)\,\omega_z\omega_x \\ \frac{1}{J_z}(J_x - J_y)\,\omega_x\omega_y \end{bmatrix} + \delta t\, J^{-1}\tau$$

b)

$$\nabla_{\omega_t} = \begin{bmatrix} 1 & \delta t\frac{1}{J_x}(J_y - J_z)\,\omega_z & \delta t\frac{1}{J_x}(J_y - J_z)\,\omega_y \\ \delta t\frac{1}{J_y}(J_z - J_x)\,\omega_z & 1 & \delta t\frac{1}{J_y}(J_z - J_x)\,\omega_x \\ \delta t\frac{1}{J_z}(J_x - J_y)\,\omega_y & \delta t\frac{1}{J_z}(J_x - J_y)\,\omega_x & 1 \end{bmatrix}$$

c)

$$\nabla_{\omega_t}(\text{sat}) = \begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{bmatrix} \qquad a_i = \begin{cases} 1 & \text{if } |\omega_t^i| \leq c \\ 0 & \text{if } |\omega_t^i| > c \end{cases}$$

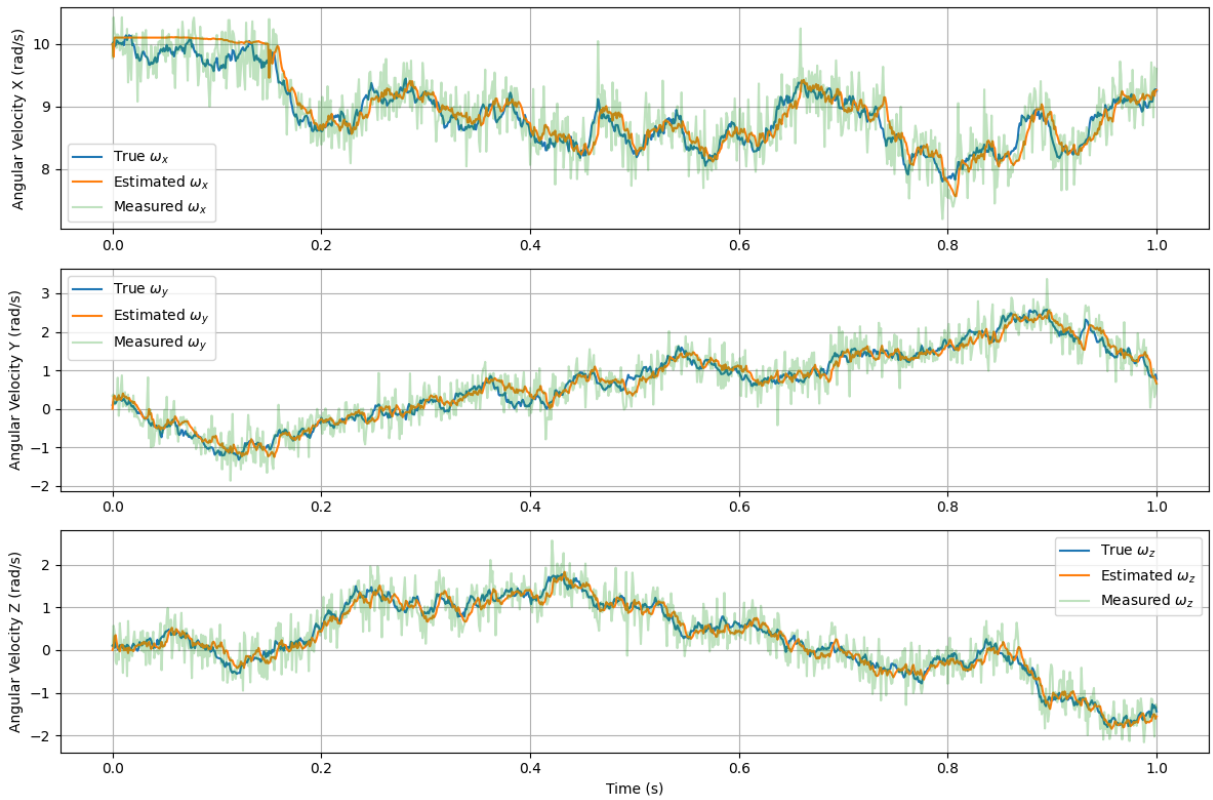
if equal to $\text{sign}(\omega_t^i)c$ then no deriv. of ω_t^i

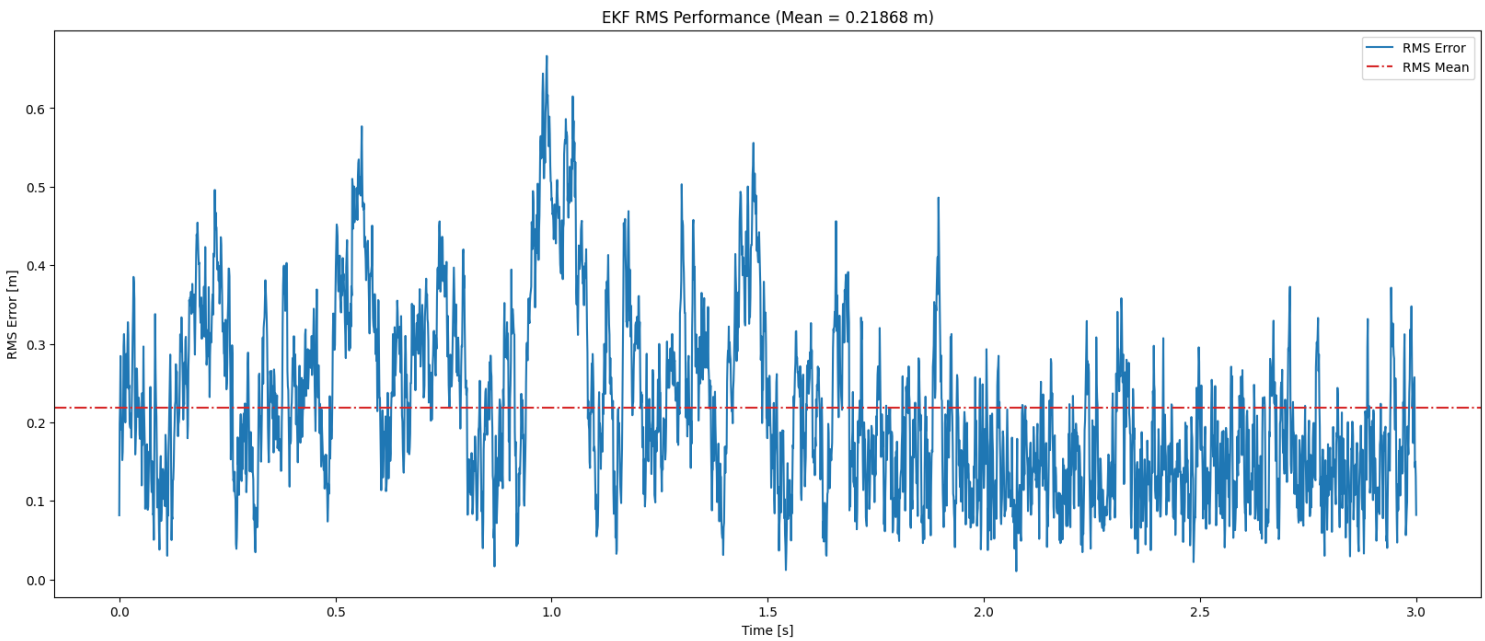
(c) Suppose the satellite gets angular velocity measurements from on-board gyroscopes with the measurement equation

$$y_t = \text{sat}(\omega_t) + v_t,$$

where $\text{sat}(\cdot)$ is the element-wise saturation function defined as follows. Let $\omega_t = [\omega_t^1 \quad \omega_t^2 \quad \omega_t^3]^T$. Then $\text{sat}(\cdot)$ returns a vector with ith element ω_t^i if $|\omega_t^i| \leq c$ for some threshold value c, and $\text{sign}(\omega_t^i)c$ if $|\omega_t^i| > c$ is above the threshold (where $\text{sign}(\cdot)$ returns 1 or -1 depending on the sign of its argument). Find the Jacobian of the measurement function.

3. [5 points] **Extended KF:** Simulate the above nonlinear discrete time system in Python, and implement an extended Kalman filter to estimate its state. Let $J = \text{diag}([1 \quad 5 \quad 5])$, $\omega_0 = [10 \quad 0.1 \quad 0.1]^T$, $c = 10$, the process noise has covariance $Q = 0.004I$, and the measurement noise has covariance $R = 0.1I$. Let your initial state estimate be $\mu_0 = [10 \quad 0 \quad 0]^T$, $\Sigma_0 = \text{diag}([1 \quad 1 \quad 1])$. Assume the input torque is zero. Attach plots showing the satellite motion, and the performance of the filter.





4. [25 points] **More Nonlinear Filters:** Consider the discrete time nonholonomic mobile robot model we saw in class

$$P_{t+1}^x = P_t^x + \delta t\, s_t \cos\Theta_t + W_t^x$$
$$P_{t+1}^y = P_t^y + \delta t\, s_t \sin\Theta_t + W_t^y$$
$$\Theta_{t+1} = \Theta_t + \delta t\, \phi_t + W_t^\theta$$

where $P_t = [P_t^x \quad P_t^y]^T$ is the robot position relative to a base station, Θ_t is its heading angle, s_t its speed input, ϕ_t its rotation rate input, and w_t is the process noise. Suppose you only get range measurements from two base stations of the form $y_t = \|p_t - p_j\| + v_t$ for station j. Assume that the base station locations are known at $(30m, -5m)$ and $(-10m, 20m)$.

Let $Q = (0.1\delta t)\mathrm{I}_3$, and $R = 0.1\mathrm{I}_2$, and let the inputs be $s_t = 1m/s$, $\phi_t = (\sin t)rad/s$. Let the initial robot state estimate be normally distributed with mean $\mu_0 = [0m, 0m, 0rad]^T$ and covariance $\Sigma_0 = .01\mathrm{I}_3$. Use $\delta t = 0.5s$ and run for 100 timesteps (i.e., 50s).

✓ (a) [6 points] Implement an EKF to estimate the robot state. Check the observability matrix for the linearized C_t, A_t Jacobians at each time step. Is the linearized system observable always, sometimes, never?

✓ (b) [6 points] Implement a UKF to estimate the robot state.

(c) [6 points] Implement a particle filter with 1000 particles to estimate the robot state.

(d) [4 points] For all the filters, plot the ground-truth trajectories and their estimated trajectories (you may find the provided simulation code useful; see `q4_simulator_script.py`). Use your code from Problem Set 2 to also plot the covariance ellipses for the position component of the state. For the particle filter, be sure to plot the samples and the sample mean. Do the filters track well? Should they be able to (why or why not)?

(e) [3 points] Note the average computation time for one loop of each of the three filters. How do they compare?
You can time the filter over the whole trajectory and divide by the trajectory length. However, you may find it more insightful to save the time of each loop into an array and then average the array. The latter approach will let you make a box-plot of the filter timing.

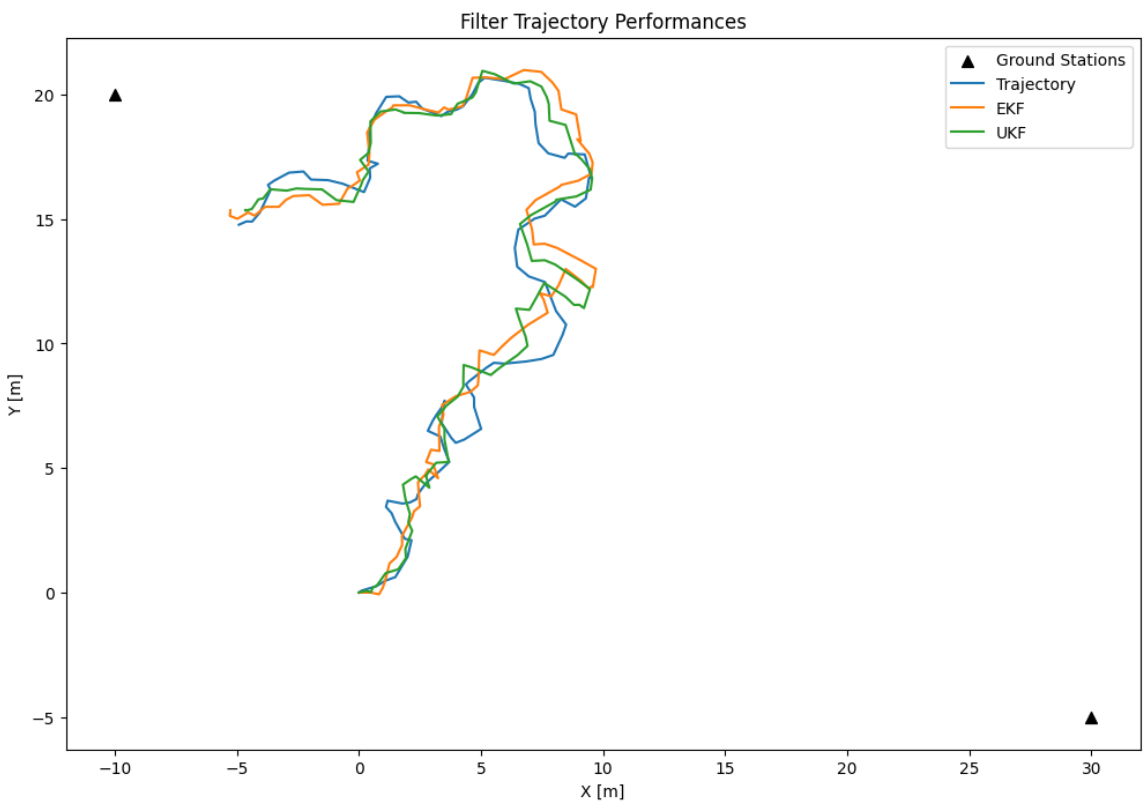
a) $x_t = \begin{bmatrix} p_t^x \\ p_t^y \\ \theta_t \end{bmatrix} \qquad u_t = \begin{bmatrix} s_t \\ \phi_t \end{bmatrix}$

$$\nabla_{x_t, u_t} f(x_t, u_t) = \begin{bmatrix} 1 & 0 & -\delta t\, s_t \sin(\theta_t) \\ 0 & 1 & \delta t\, s_t \cos(\theta_t) \\ 0 & 0 & 1 \end{bmatrix} = A$$

$$y_t = \sqrt{(p_t^x - x_j)^2 + (p_t^y - y_t)^2}$$

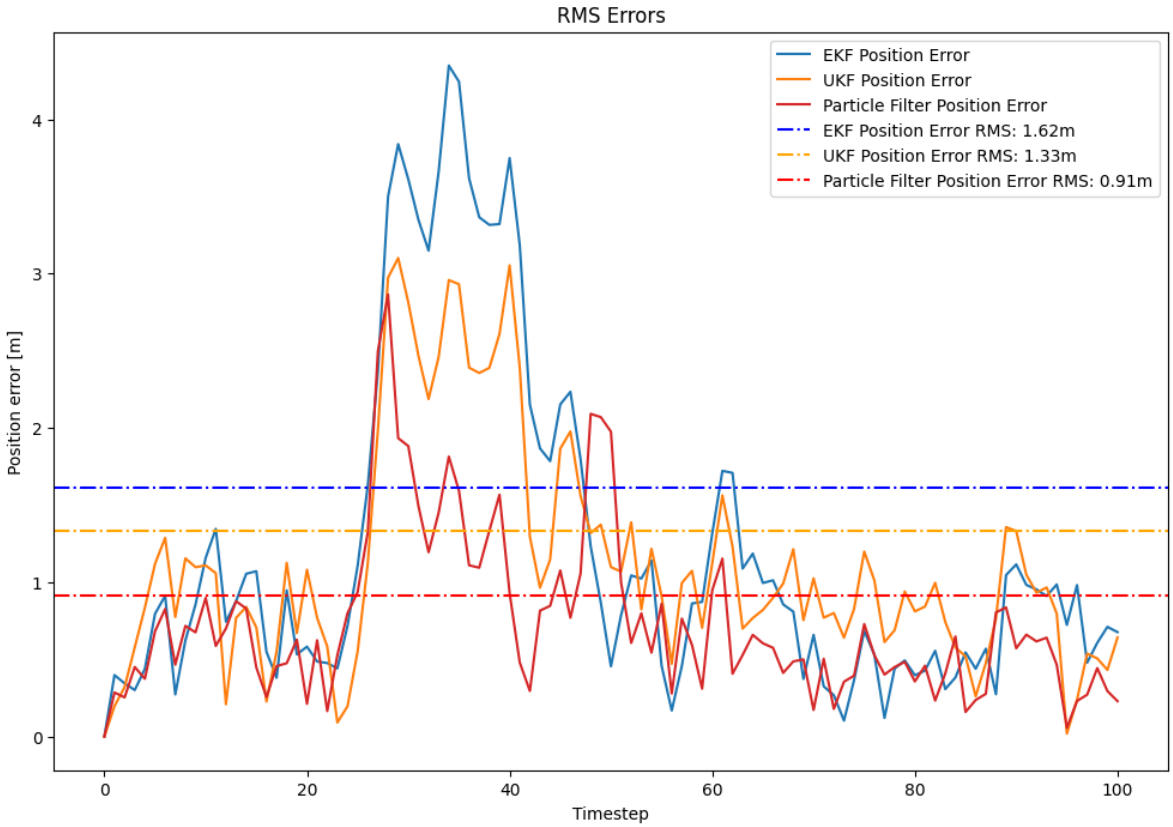
$$\nabla_{x_t} g(x_t) = \begin{bmatrix} \frac{1}{2}\left((p_t^x - x_1)^2 + (p_t^y - y_1)^2\right)^{-\frac{1}{2}}(2)(p_t^x - x_1) & \frac{1}{2}\left((p_t^x - x_1)^2 + (p_t^y - y_1)^2\right)^{-\frac{1}{2}}(2)(p_t^y - y_1) \\ (\ldots \text{ w/ } x_2, y_2) & (\ldots \text{ w/ } x_2, y_2) \end{bmatrix}$$

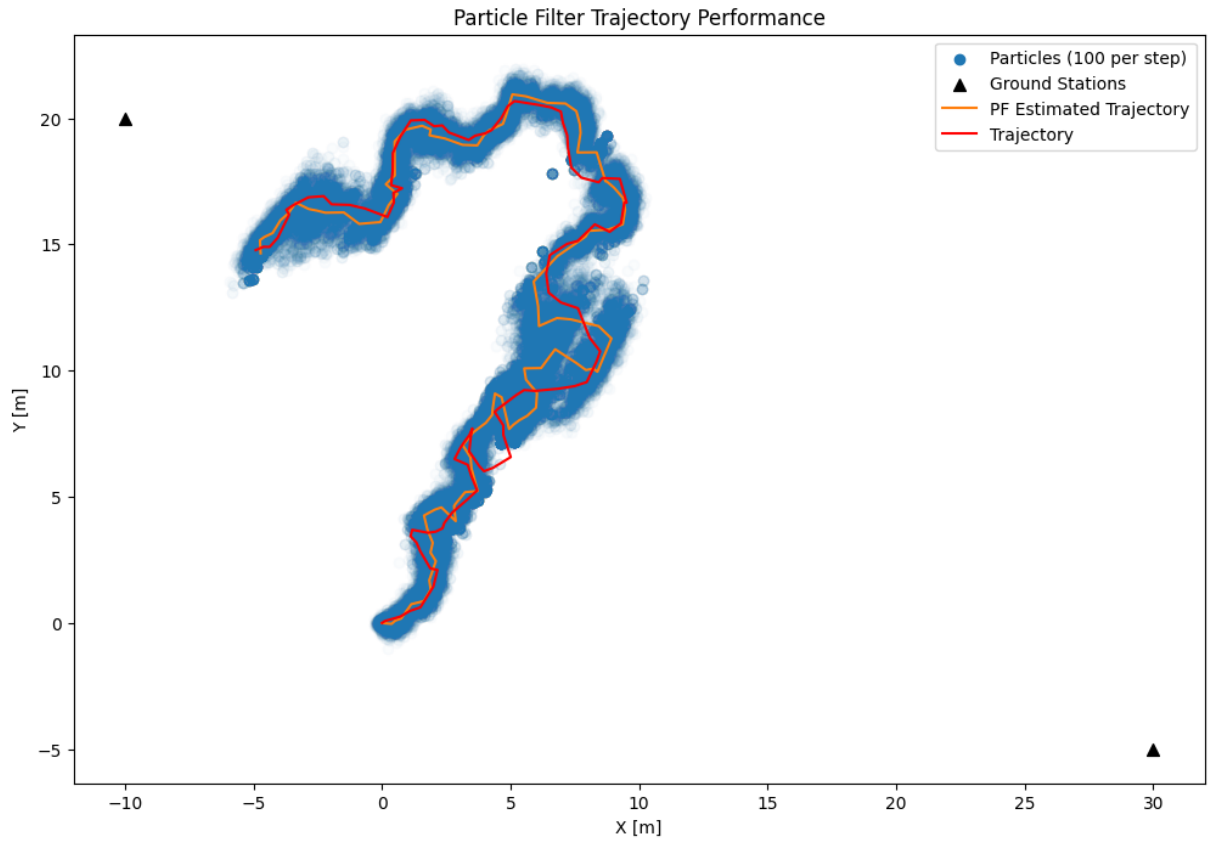
$$= \begin{bmatrix} \frac{p_t^x - x_1}{\sqrt{(p_t^x - x_j)^2 + (p_t^y - y_t)^2}} & \frac{p_t^x - y_1}{\sqrt{(p_t^x - x_j)^2 + (p_t^y - y_t)^2}} & 0 \\ \frac{p_t^x - x_2}{\sqrt{(p_t^x - x_j)^2 + (p_t^y - y_t)^2}} & \frac{p_t^x - y_2}{\sqrt{(p_t^x - x_j)^2 + (p_t^y - y_t)^2}} & 0 \end{bmatrix}$$

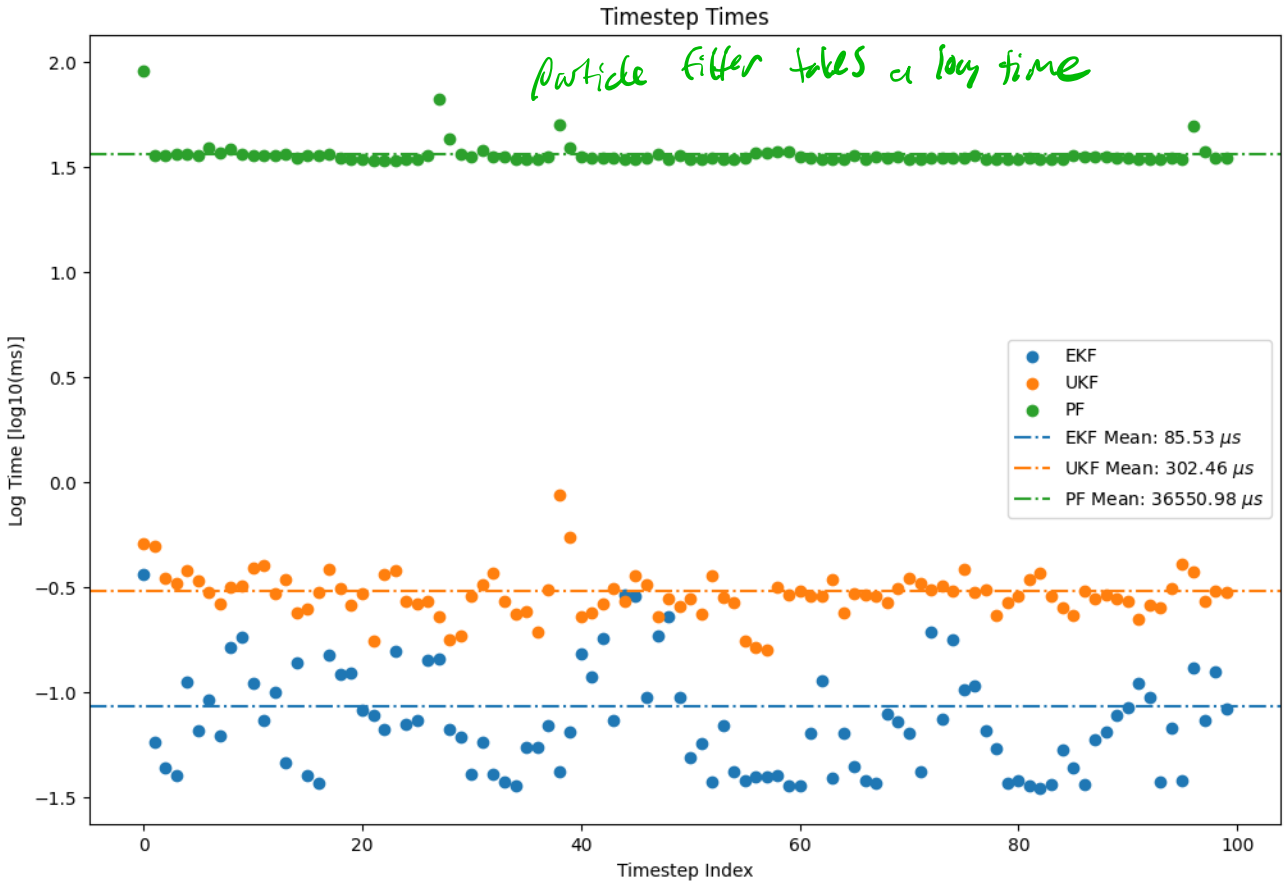


Fully observable. At all timesteps in code, the O matrix is full rank

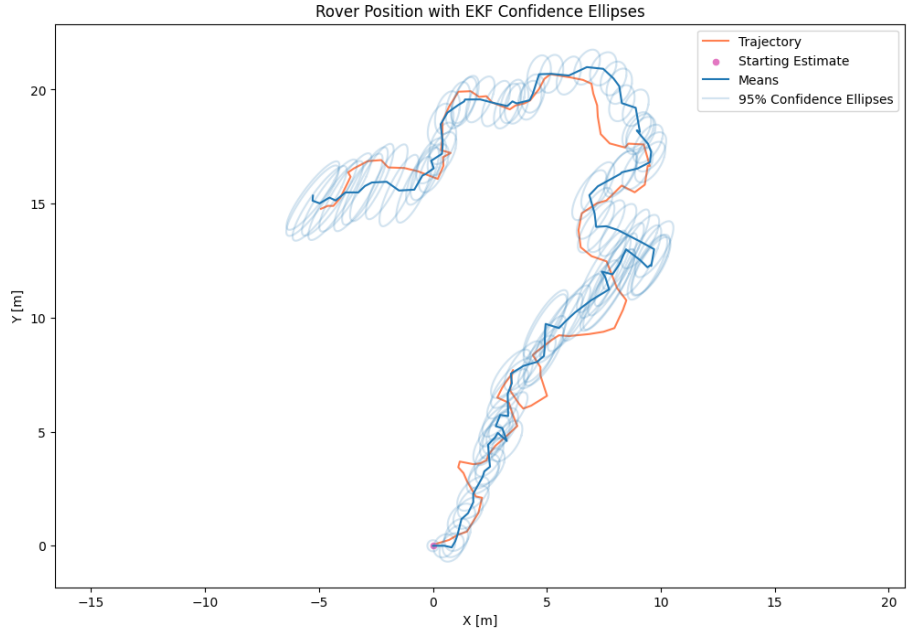
All 3 track relatively well, with the particle filter having the smallest RMS error. The should be able to track well because of their absolute measurements.

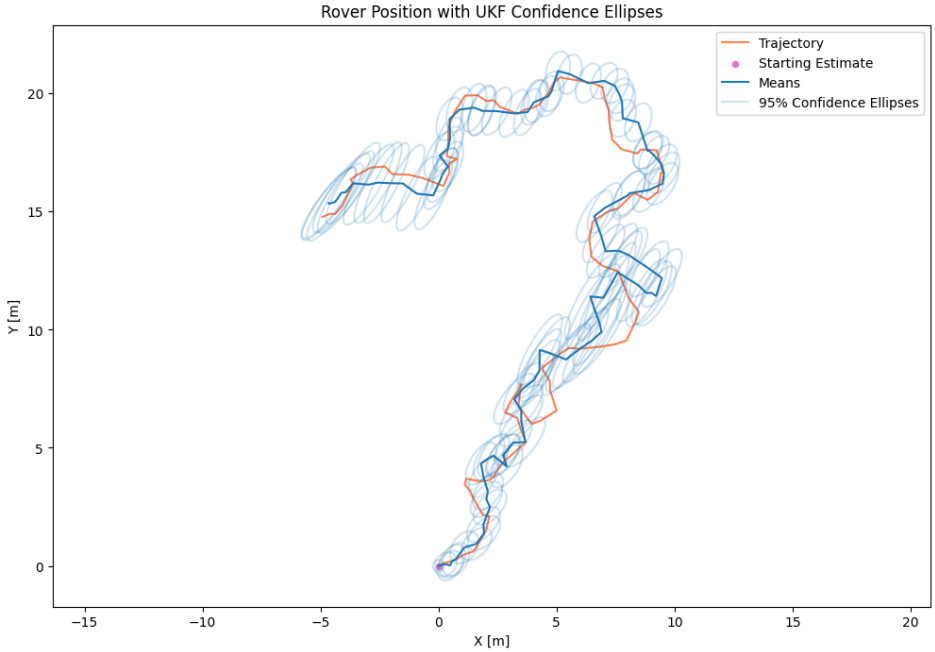


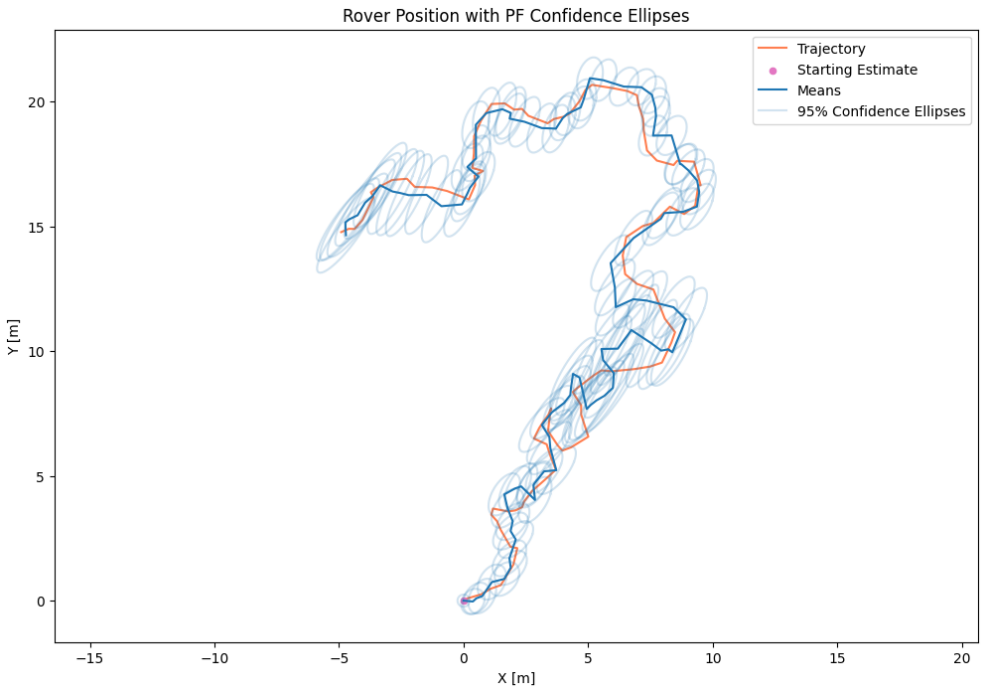




generally, longer execution time is correlated with better performance.


Rover Position with EKF Confidence Ellipses
Trajectory
Starting Estimate
Means
95% Confidence Ellipses
Y [m]
X [m]
20
15
10
5
0
−15
−10
−5
0
5
10
15
20


Rover Position with UKF Confidence Ellipses
Trajectory
Starting Estimate
Means
95% Confidence Ellipses
Y [m]
X [m]
20
15
10
5
0
−15
−10
−5
0
5
10
15
20


Rover Position with PF Confidence Ellipses
Trajectory
Starting Estimate
Means
95% Confidence Ellipses
Y [m]
X [m]
20
15
10
5
0
−15
−10
−5
0
5
10
15
20